Supplementary Material

I. PARAMETER-DEPENDENT ALGORITHM IMPOSING THE NEGATIVENESS ON THE TRANSVERSAL LYAPUNOV EXPONENT

Suppose we have n_p scalar time series measured for different values of parameter p: $\{p_j\}_{j=1,..,n_p}$. We assume that all time series have the same length N and the same embedding delay time can be used for each of them. We will incorporate all these time series into the training procedure so that the NGRC will be able to replicate the system behavior for parameter values not included in the training set. Moreover, we will extend this algorithm by imposing a new requirement that the Transversal Lyapunov Exponent (TLE) will be equal to some negative value for all parameter values involved in the learning. The parameter-dependent algorithm was already presented in Ref.[23]. Here we will supplement it by requirement for TLE. The requirement for TLE was already proposed in Ref.[29] for the case of single parameter value.

The resulting matrix equation for optimization problem with a single parameter value reads (Ref.[29]):

$$\mathbf{W}_\lambda A_\lambda = \mathbf{b}_\lambda, \tag{1}$$

with

$$\mathbf{W}_\lambda = \left[\mathbf{W}, W_{d+1}\right], \tag{2a}$$

$$\mathbf{b}_\lambda = \left[\mathbf{b}, \lambda_\perp\right], \tag{2b}$$

$$A_\lambda = \begin{bmatrix} A & \mathbf{C}^T \\ \mathbf{C} & 0 \end{bmatrix}. \tag{2c}$$

See also Eqs. (28), (27), (31), and (29) in manuscript.

Here $A = \widetilde{O}\widetilde{O}^T + \beta I$, and $\mathbf{b} = \widetilde{D}\widetilde{O}^T$. W_{d+1} is the Lagrange multiplier, and $\lambda_\perp$ is the required TLE. The components of the row-vector $\mathbf{C}$ are given by

$$C_j = \frac{1}{N}\sum_{i=1}^{N} \widetilde{O}'_{ji}, \quad j = 1,..,d. \tag{3}$$

Here the prime denotes the partial derivative with respect to the first argument (see Eq.(22) in manuscript).

In the form given by Eqs.(1), (2), the optimization problem will be solved for a single fixed parameter value. However, if we wish to fit the output vector $\mathbf{W}_\lambda$ to several parameter values simultaneously, we have to expand the system (1), (2). In the expanded form for two parameter values involved ($n_p = 2$), the above expressions are converted to the following ones:

$$\mathbf{W}_\lambda = \left[\mathbf{W}, W_{d+1}, W_{d+2}\right], \tag{4a}$$

$$\mathbf{b}_\lambda = \left[\mathbf{b}, \lambda_\perp, \lambda_\perp\right], \tag{4b}$$

$$A_\lambda = \begin{bmatrix} A & \mathbf{C}^{(1)T} & \mathbf{C}^{(2)T} \\ \mathbf{C}^{(1)} & 0 & 0 \\ \mathbf{C}^{(2)} & 0 & 0 \end{bmatrix}. \tag{4c}$$

The matrix A and row-vector $\mathbf{b}$ now become the sums:

$$A = \widetilde{O}_1\widetilde{O}_1^T + \widetilde{O}_2\widetilde{O}_2^T, \tag{5a}$$

$$\mathbf{b} = \widetilde{D}_1\widetilde{O}_1^T + \widetilde{D}_2\widetilde{O}_2^T. \tag{5b}$$

The solution of the two-parameters optimization problem can be written as:

$$\mathbf{W}_\lambda = \mathbf{b}_\lambda (A_\lambda + \beta I)^{-1}, \tag{6}$$

where β is a small Tikhonov ridge regularization parameter, and I is a unity matrix of $(d+2)\times(d+2)$ size. The lower (upper) indices under (over) matrices and vectors (e.g. $\widetilde{O}_1$, $\widetilde{D}_1$, $\mathbf{C}^{(1)}$) imply that these quantities were calculated at the corresponding values (e.g. $p_{j=1}$) of the parameter. Note that the feature vectors $\widetilde{O}_j$ are composed of the Tschebyshev polynomials, in which the corresponding parameter value p_j is incorporated as a variable (see Appendix in Ref.[23]).

If one needs to involve more parameter values, the generalization of equations (4), (5),(6) is straightforward.

In the vector $\mathbf{W}_\lambda$, last n_p entries are Lagrange multipliers; in the vector $\mathbf{b}_\lambda$, last n_p entries are equal to the imposed value of TLE, $\lambda_\perp$, and in the matrix A_λ, in the right lower corner there is located the zero matrix of dimension $n_p \times n_p$.

When integrating the hybrid NGRC-Ch system, in the vector $\mathbf{W}_\lambda$ the last n_p entries must be omitted, i.e. only the sub-vector $\mathbf{W}$ is used. The hybrid NGRC-Ch system now reads as follows:

$$d\nu/dt = \mathbf{W}\mathbf{O}[\nu(t), \nu(t-\tau), .., \nu(t-n\tau), p]. \tag{7}$$

Here the output vector $\mathbf{W}$ is computed by invoking n_p values of system parameters p_j ($j = 1,..,n_p$), and the feature vector contains some value of parameter p that is now treated as an additional argument (together with $\nu(t), \nu(t-\tau), .., \nu(t-n\tau)$). The value p has to be in the parameter interval filled by parameter values p_j, ($j = 1,..,n_p$), used in the learning phase.

II. COMPUTING BIFURCATION DIAGRAM WITH HYBRID NGRC-CH ALGORITHM

First, we collect time series of variable $x(t)$ at the given values of parameter, p_j, ($j = 1,..,n_p$). With these series, we perform the learning, i.e. compute the output vector $\mathbf{W}_\lambda$, following the algorithm described above. Next, we gradually move the parameter in the interval $[p_{\min}, p_{\max}]$, and integrate the Eq.(7) for each value of p. We thus get time series of $\nu(t)$ for each scanned p.

For each time series we compute the set of existing time intervals between subsequent spikes. These computations are performed by the same techniques as those for the time series of original HR neuron system.

We thus get the bifurcation diagram computed from the hybrid NGRC-Ch system, which was learned from the original HR system at several parameter values.